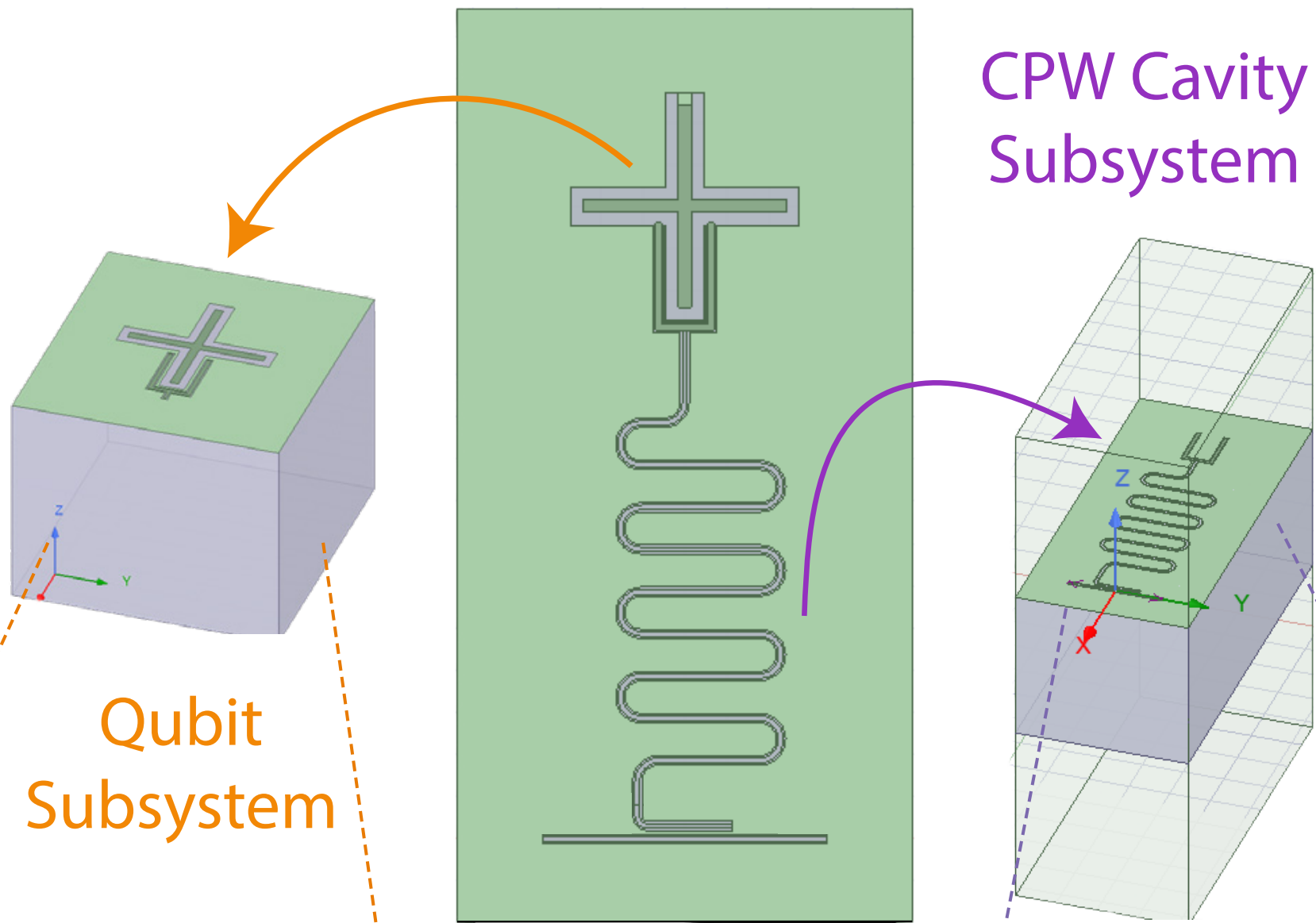




CPW Cavity
Subsystem

Qubit
Subsystem

α (anharmonicity)
 $\approx -E_C$ from Q3D

f_{qubit} (qubit frequency)
 $\approx \sqrt{(8 E_J E_C)} - E_C$ from Q3D

f_{res} (resonator freq.)
HFSS eigenmode solver

κ (resonator linewidth)
HFSS via 50 Ω port loss